HEALTH

How do city lights affect sleep?

Most people now live under a night sky that never fully darkens. The body's clock reads that light as a signal.

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An unmade bed and a window onto city lights. · AI-generated image

Most people now try to sleep under a sky that never truly darkens and whose stars are fading. A 2016 mapping found that over 80 percent of the world's population – and over 99 percent in the US and Europe – live under light-polluted skies, hiding the Milky Way from a third of humanity. [1] From volunteers' reports between 2011 and 2022, Kyba and colleagues calculated that the sky is brightening by about 9.6 percent a year. [2]

Our bodies read light as a "time signal." Evening light, especially blue light, delays the release of melatonin, the hormone preparing us for sleep, and the white LEDs now common in street lighting emit a large share of their light in that blue range. A study of nearly 16,000 Americans found that people in brighter areas tended to go to bed later, sleep less, and feel less satisfied with their rest. [3] The link is an association, not proof; the light that reaches a bedroom also depends on curtains and screens. Cities that want darker nights have simple tools: downward-pointing shielded lamps, warmer hues, and dimming in the late hours.

Sources

1. Falchi et al. – The new world atlas of artificial night sky brightness (Science Advances, 2016)
https://doi.org/10.1126/sciadv.1600377

2. Kyba et al. – Citizen scientists report global rapid reductions in the visibility of stars from 2011 to 2022 (Science, 2023)
https://doi.org/10.1126/science.abq7781

3. Ohayon & Milesi – Artificial Outdoor Nighttime Lights Associate with Altered Sleep Behavior in the American General Population (Sleep, 2016)
https://doi.org/10.5665/sleep.5842